

CSE 401: Numerical Analysis, Simulation and Modeling



Nafis Tahmid

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CSE'19 Batch

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CSE401

Numerical Analysis, Simulation and Modeling

3 Credit Hour Course

Intended For Level 4 Term 1 Students

Prerequisite:

Introduction to Numerical Analysis, approximations, round-off errors, truncation errors; Visualization and plotting; Root finding: bisection method, false position method, Newton-Raphson method, Bairstow's method; Solution of systems of equations: Gauss elimination method, Gauss-Jordan elimination method, LU decomposition; Eigenvalue decomposition: power method, QR method; Optimization: golden-section search, Newton's method, gradient methods, constrained optimization; Curve fitting, interpolation and approximation: least squares regression, linear interpolation, Lagrange polynomial interpolation, Newton's polynomial interpolation, spline interpolation; Numerical integration and differentiation: Newton-Cotes integration, trapezoidal rule, Simpson's rule, Romberg's integration, Richardson's extrapolation; Solution of ordinary differential equations: Euler's method, Runge-Kutta methods, finite difference methods; Modeling with linear and differential equations; Introduction to simulation and modeling, Discrete event simulation models; Steps in a simulation study; Model validation and verification; Random number generation; Monte Carlo methods: Metropolis-Hastings algorithm

- Continuous Assessment
- Anonymous Feedback
- Textbook:
 - Numerical Part
 - Simulation: Simulation Modeling and Analysis (5th Edition) by Averill M. Law

CSE 401: Numerical Analysis, Simulation and Modeling

Introduction, Approximations & Errors

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Last Compiled: June 21, 2026

*Everything you compute over the next few weeks will be wrong.
This course spends an uncomfortable amount of time asking how wrong, and
whether wrong is still useful.*

The big idea

Engineering questions become *mathematical models*. A lot of useful models cannot be solved with pencil-and-paper formulas, so we solve them *approximately* — that approximation is numerical methods.

The big idea

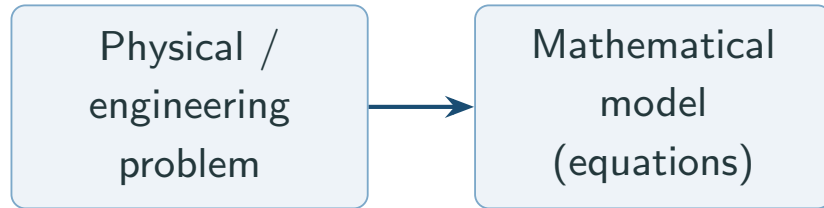
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Calculus promises exact answers; numerical methods are what you do when calculus stops returning your calls.

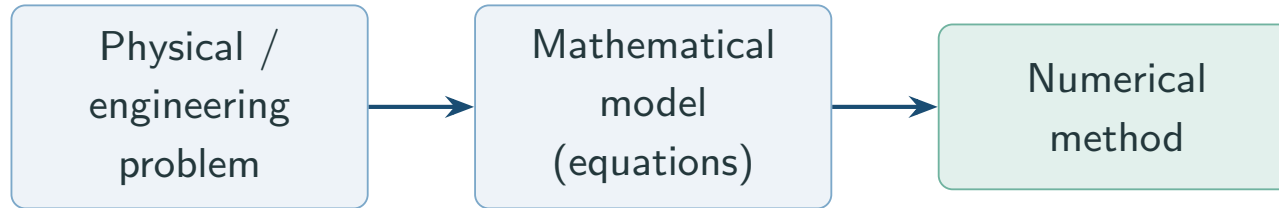
From a physical problem to a number

Physical /
engineering
problem

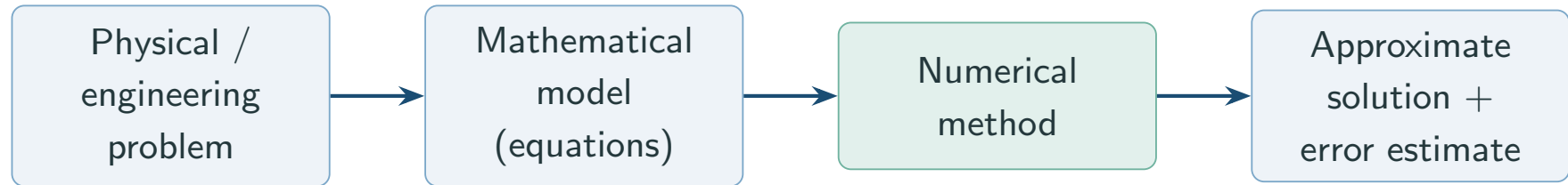
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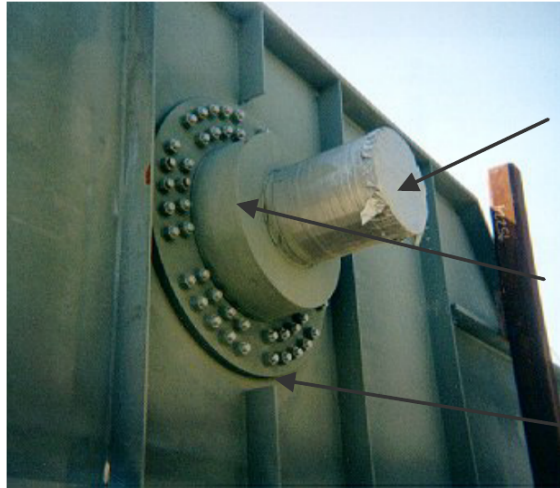


Bascule-Bridge

A common type of movable drawbridge that opens to allow boat traffic to pass underneath



The bascule-bridge trunnion



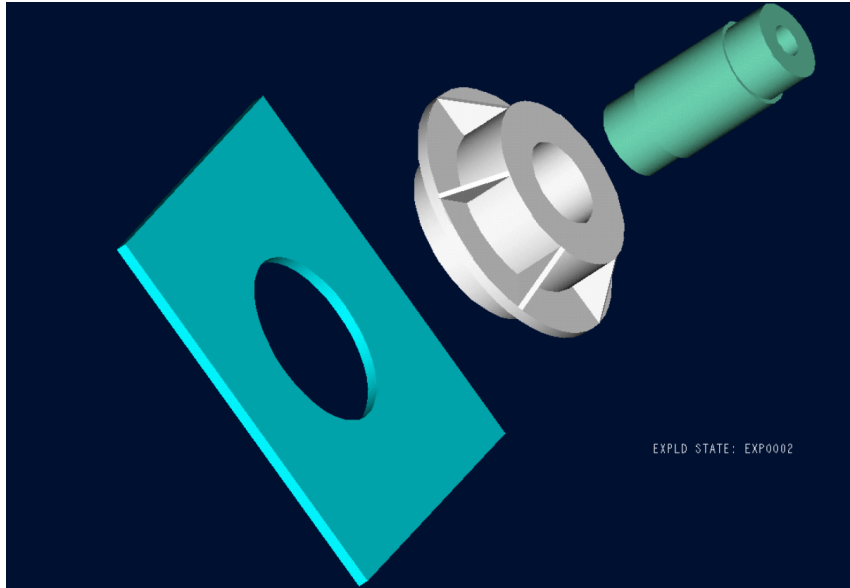
Trunnion

Hub

Girder

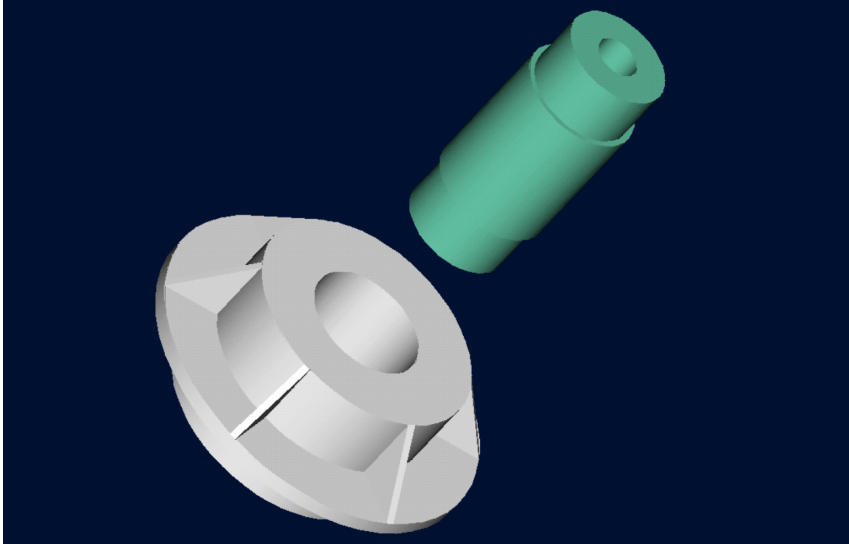
A hollow steel *trunnion* is shrink-fit into a *hub*, then into the bridge *girder*.

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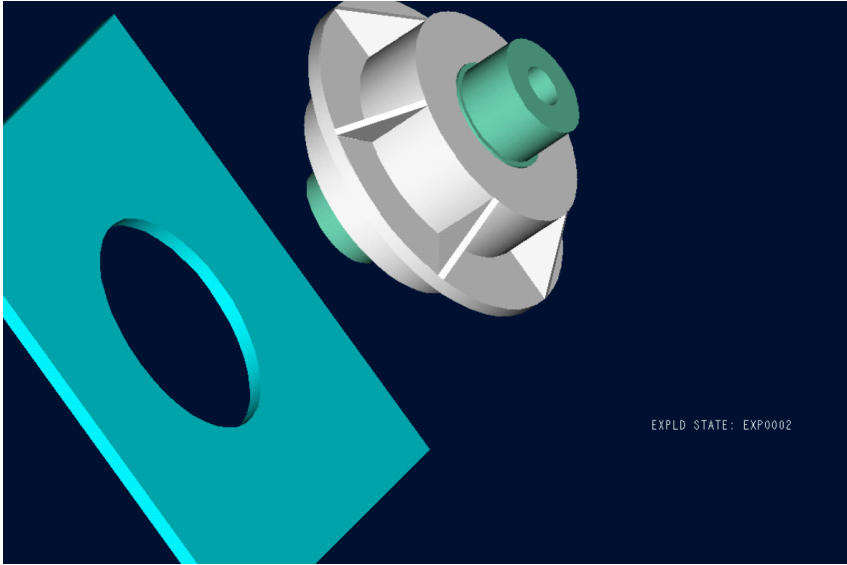
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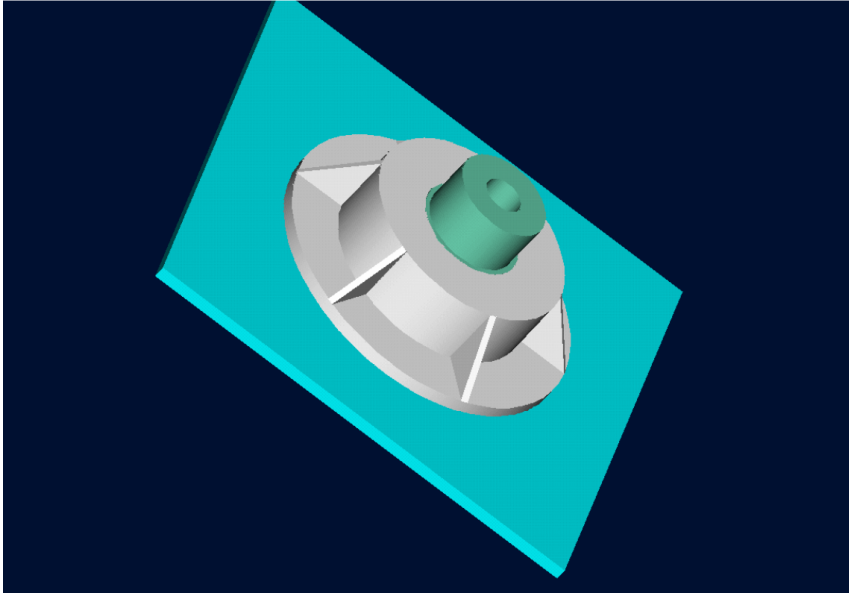
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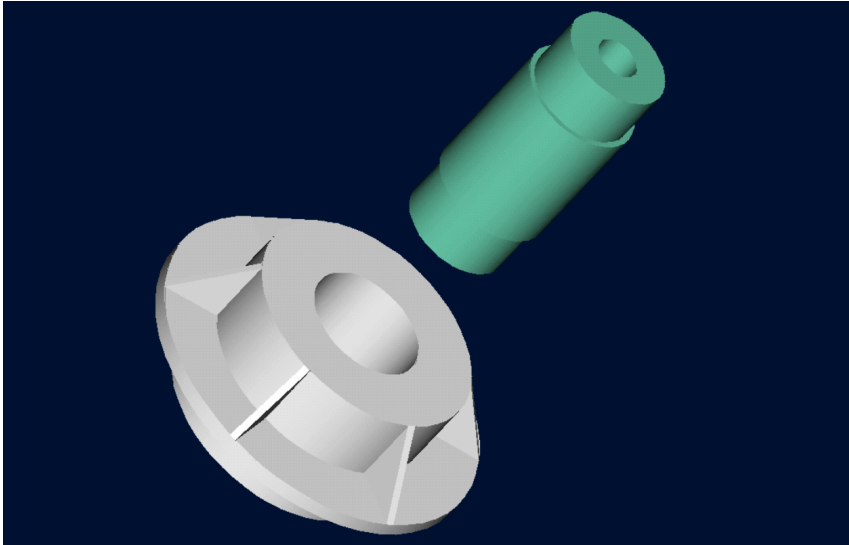
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- Trunnion-hub chilled and then pressed into the girder, where they warm-up to a tight fit.

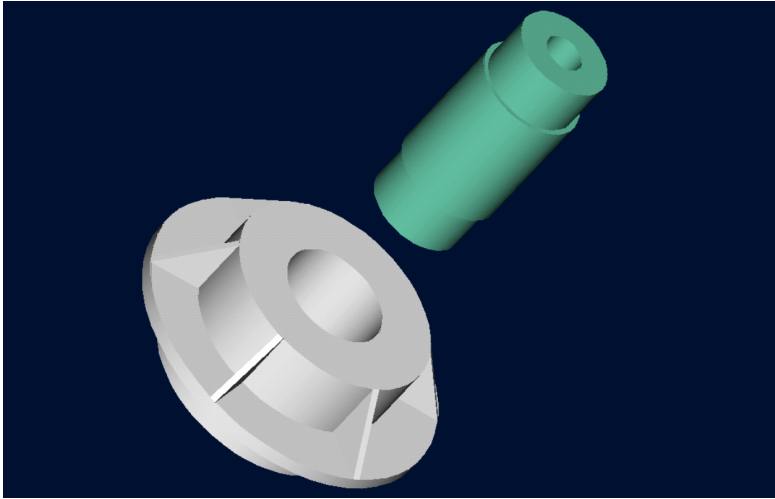
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- On warming it expands $\Rightarrow$ a tight fit.
- Trunnion-hub chilled and then pressed into the girder, where they warm-up to a tight fit.

Florida, 1995: the trunnion *got stuck* mid-insertion. A replacement trunnion + hub would cost \$50,000; with delays, a six-figure loss. *The trunnion was pulled out before it stuck permanently.*



Trunnion OD = 12.363'', hub ID = 12.358'', required diametric clearance = 0.01''. Dry-ice/alcohol bath at -108°F ; room temperature 80°F .

Required contraction

$$12.363 - 12.358 + 0.01 = 0.015''$$

Question. Will immersion in dry-ice/alcohol contract the trunnion by at least 0.015''?

First model — constant expansion coefficient

Using a single (room-temperature) coefficient α :

$$\Delta D = D \alpha \Delta T, \quad \Delta T = T_{\text{fluid}} - T_{\text{room}} = -108 - 80 = -188^\circ\text{F}.$$

Example (plug in the numbers). With $D = 12.363''$ and $\alpha = 6.47 \times 10^{-6} \frac{\text{in}}{\text{in } ^\circ\text{F}}$:

$$\Delta D = (12.363)(6.47 \times 10^{-6})(-188) = -0.01504''.$$

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Looks fine on paper

Predicted contraction $0.01504'' > 0.015''$ required — so the bath *should* work... yet the trunnion got stuck. What went wrong: a bad model, or a calculation error?

But α varies with T

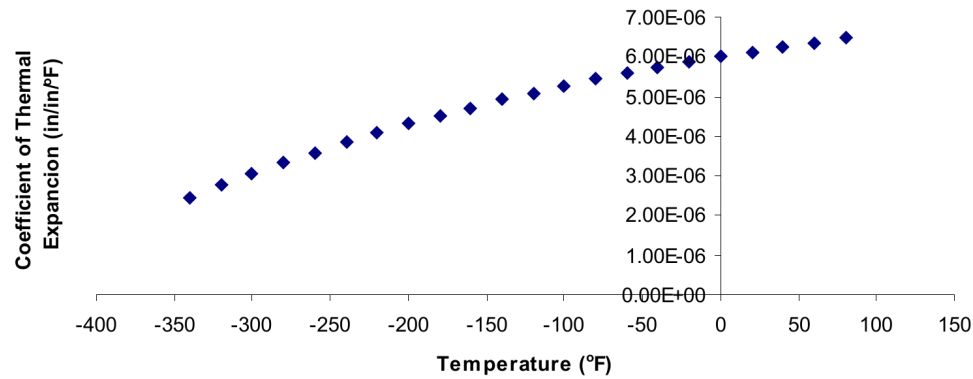


Table 1 (abridged).

T (°F)	α ($\mu\text{in/in}/^\circ\text{F}$)
80	6.47
40	6.24
0	6.00
−40	5.72
−100	5.28
−160	4.72
−220	4.08
−300	3.07
−340	2.45

Because $\alpha = \alpha(T)$, the contraction is an *integral*, not a product:

$$\Delta D = D \int_{T_{\text{room}}}^{T_{\text{fluid}}} \alpha(T) dT.$$

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We assume a second-order polynomial.

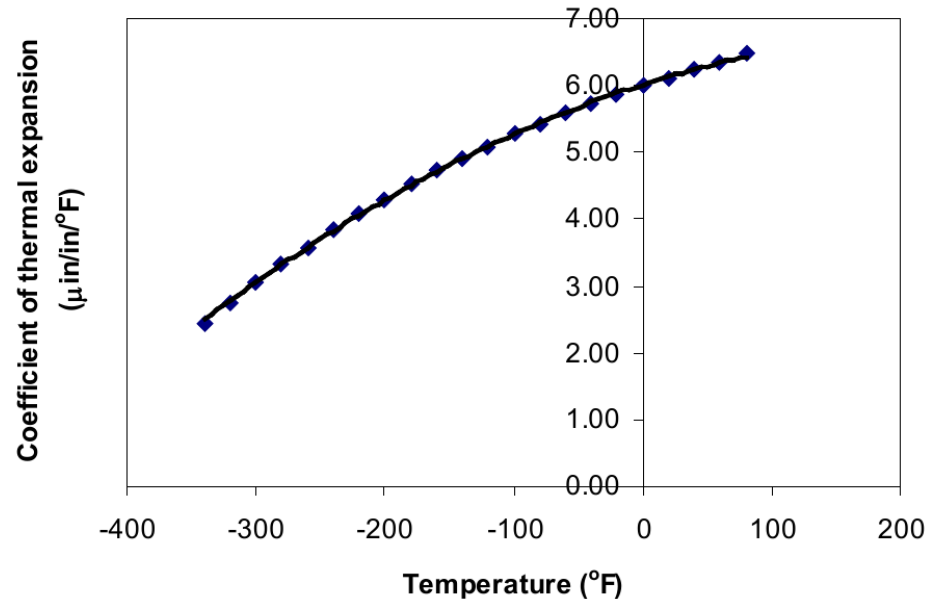
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We assume a second-order polynomial.

$$\alpha(T) = a_0 + a_1 T + a_2 T^2, \quad a_0 = 6.0150 \times 10^{-6}$$
$$a_1 = 6.1946 \times 10^{-9}$$
$$a_2 = -1.2278 \times 10^{-11}$$



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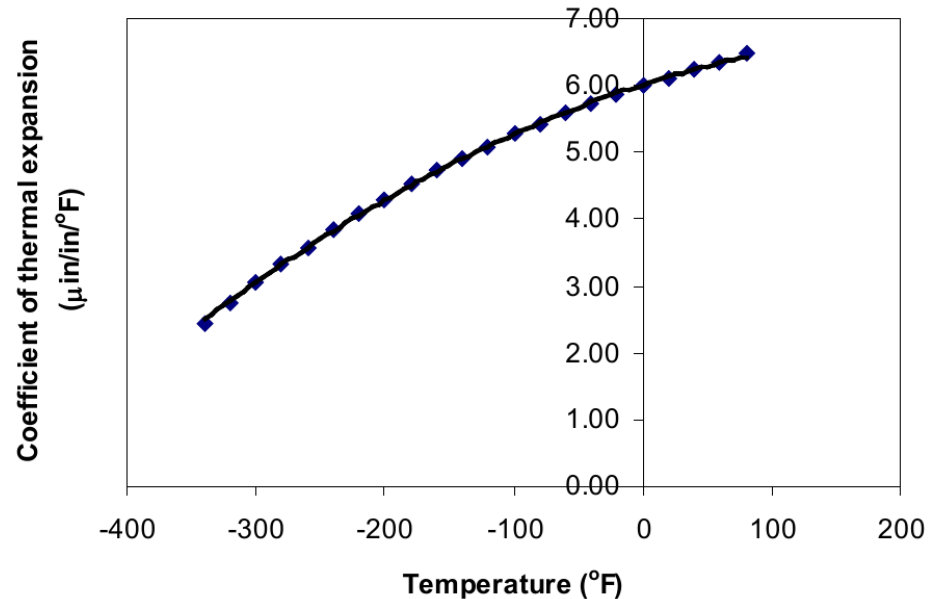
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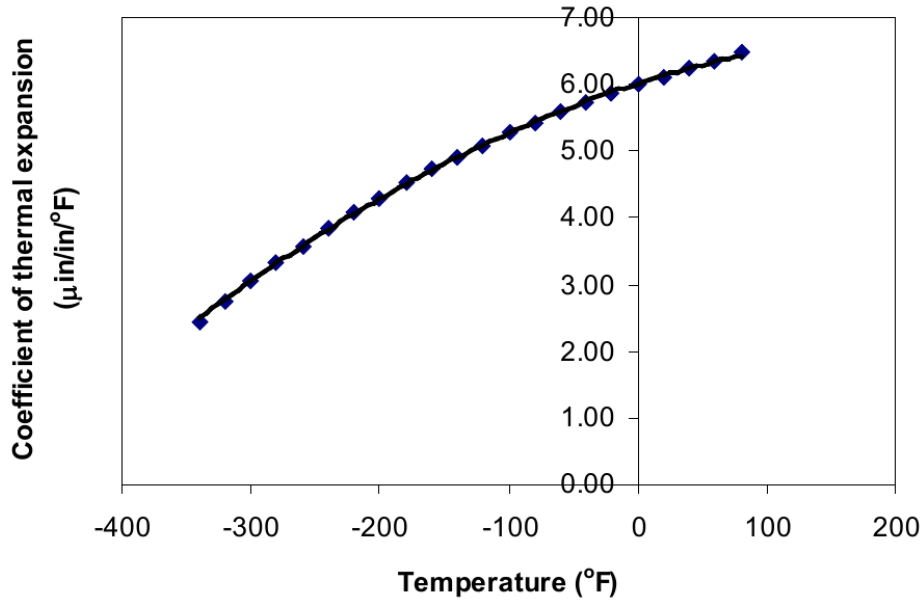


Connecting
the dots



Interpolating a
polynomial of order
1 through 2 points



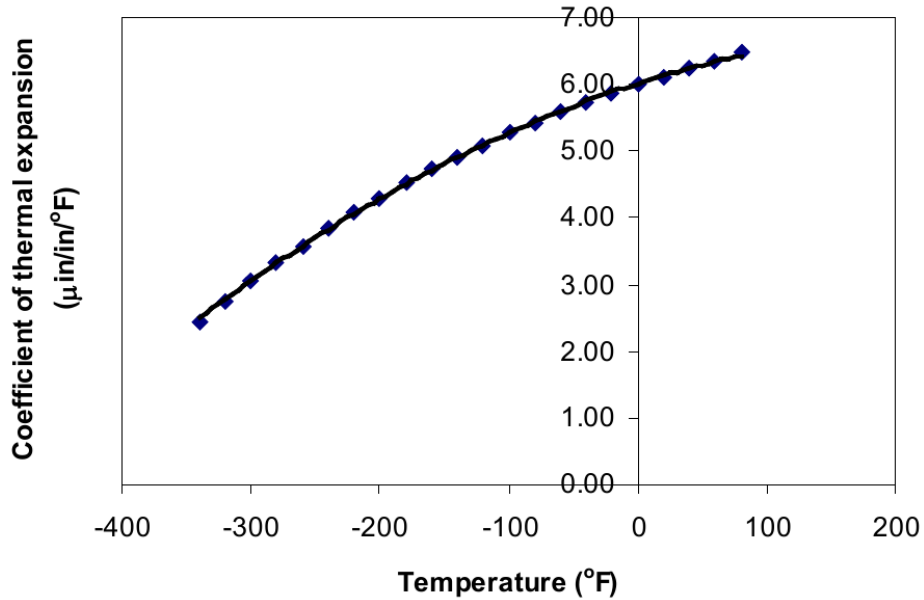


Carrying out the integral with the fitted $\alpha(T)$:

$$\Delta D = -0.013689''.$$

Getting stuck was predictable

Actual contraction $0.013689'' < 0.015''$ (required).
The trunnion didn't shrink enough, so got stuck.
The constant- α formula was wrong.



Carrying out the integral with the fitted $\alpha(T)$:

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The constant- α formula was wrong.

We need good **models** and **tools** to solve them, and we need to know **how wrong** the answers are, so we can decide if they are good enough.

 : I don't like **math**,
I'm just going to college
to choose "**IT**" **major**.

 "**IT**" **Major** : Welcome to
Calculus, Linear Algebra,
Mathematical logic, Statistics,
Numerical Methods, Discrete
mathematics.

workflow $\rightarrow$ data থেকে mathematical model বের করি। Model এ error introduce হতে পারে। Plus analytical solution বের করা possible না হলে numerical approach নেই।

Numerical answers are *approximate*, so error is unavoidable. To handle it we:

1. **identify** where error comes from,

2. **quantify** it, and $\rightarrow$ goal: acceptability $\rightarrow$ যেটা method follow করলাম সেটা acceptable কিনা

3. **minimise** it to our needs.

$\hookrightarrow$ approach এর কোথাও change করা হলে error minimise হবে বের করতে হবে।

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Step (2): quantify

We need numbers that say *how wrong* an approximation is and, crucially, numbers we can compute even when the true answer is unknown.

True ans জ্ঞানে approximate করতে হবে।

Definition

$$E_t = \text{True value} - \text{Approximate value.}$$

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$\rightarrow f'(x) = 7 \times 0.5 e^{0.5x}$

Example 1. Approximate $f'(2)$ for $f(x) = 7e^{0.5x}$ by a forward difference with $h = 0.3$:

$h \rightarrow 0$ $\nearrow$ $f'(x) \approx \frac{f(x+h) - f(x)}{h} \Rightarrow f'(2) \approx \frac{7e^{0.5(2.3)} - 7e^{0.5(2)}}{0.3} = \frac{22.107 - 19.028}{0.3} = 10.265.$

Exact value: $f'(x) = \underline{3.5e^{0.5x}} \Rightarrow f'(2) = 3.5e^1 = \underline{9.5140}$. Hence

actual ans দেব করতে পারি

$$E_t = 9.5140 - 10.265 = -0.75061.$$

The minus sign: we estimated too high.

Magnitude alone is misleading

An error of $E_t = -0.75061$ *sounds* small.

Same problem, rescaled. If instead $f(x) = 7 \times 10^{-6} e^{0.5x}$, every value shrinks by 10^{-6} , so

$$E_t = -0.75061 \times 10^{-6}.$$

The problem is *identical* (same x , same h), yet the raw error looks 10^6 times “better”.

We need an error *relative to the size of the answer*.

Relative true error ϵ_t

True value বা truth বোঝানো

Definition

$$\epsilon_t = \frac{\text{True error}}{\text{True value}} = \frac{E_t}{\text{True value}}.$$

Example 2. For Example 1,

$$\epsilon_t = \frac{-0.75061}{9.5140} = -0.078895 = -7.8895\%.$$

As an *absolute relative true error*: $|\epsilon_t| = 0.078895 = 7.8895\%.$

Approximate error E_a — when the truth is unknown

We rarely know the true value (if we did, why approximate?).

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We try to find solutions *iteratively* until they *converge* (*good enough*).

- prev step এর সাথে diff কম
- * # iterations cap reached.

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Example 3. Same derivative, two step sizes:

(let) $\swarrow$ 3rd step $h = 0.3 : f'(2) \approx 10.265,$ $\nwarrow$ 4th step $h = 0.15 : f'(2) \approx \frac{7e^{0.5(2.15)} - 7e^1}{0.15} = 9.8799.$

$$E_a = 9.8799 - 10.265 = \underline{-0.38474}.$$

approximate error

Relative approximate error ϵ_a

Definition

$$\epsilon_a = \frac{\text{Approximate error}}{\text{Present approximation}} = \frac{E_a}{\text{Present approximation}}.$$

Example 4. From Example 3,

$$\epsilon_a = \frac{-0.38474}{9.8799} = -0.038942 = -3.8942\%, \quad |\epsilon_a| = 3.8942\%.$$

absolute relative approximate error

Relative approximate error ϵ_a

present approx to truth $\frac{E_a}{\text{Present approx}}$,
→ definitely better than prev approx.

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absolute value

ϵ_a is computable *without the true value*, this is the quantity we actually monitor.

Two ways to stop an iteration

(i) Pre-specified tolerance ϵ_s *threshold*

Stop when $|\epsilon_a| \leq \epsilon_s$. Choose ϵ_s to match the accuracy you need.

(ii) Pre-specified significant digits m

To trust at least m significant digits in the answer, iterate until

$$|\epsilon_a| \leq 0.5 \times 10^{2-m} \% .$$



$$\hat{x} = 5.342678$$

বড়দুই পর্যন্ত digit correct আছে confidently
বলতে পারি।

$$x = \underline{\underline{5.34}} \dots$$

$$\hat{x} \approx$$

প্রথম 3 টি exactly
match করে; পরের
যুলো sure না 23

Example 5

The Maclaurin series: For a function $f(x)$ with derivatives at $x = 0$:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

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Estimate $e^{0.7}$ from the Maclaurin series $e^x = 1 + x + \frac{x^2}{2!} + \cdots$

$$f(x) = e^x$$

$$f^{(n)}(x) = e^x$$

$$f^{(n)}(0) = 1$$

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$$|\epsilon_a| = \left| \frac{\overset{\text{present}}{2.0136} - \overset{\text{prev}}{2.0122}}{2.0136} \right| \times 100\% = 0.069527\%.$$

$$|\epsilon_a| \leq 0.5 \times 10^{2-m} \% \rightarrow \text{at least } m \text{ significant bits correct}$$

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$$|\epsilon_a| = \left| \frac{2.0136 - 2.0122}{2.0136} \right| \times 100\% = 0.069527\%.$$

Test against $m = 2$: $0.5 \times 10^{2-2}\% = 0.5 \times 10^0\% = \underline{0.5\%}$.

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Conclusion

Since $0.069527\% \leq 0.5\%$, at least 2 significant digits are correct if we use 6 terms

What is a significant digit?

Significant digits report *how much you trust* a number.

Example. “The county population is **1 million**”, only *one* significant digit (1).

The exact count 1,079,587 has seven.

→ ১ম থেকে বেশি। কত বেশি বলা থাকেনা।

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Scientific notation makes the difference explicit:

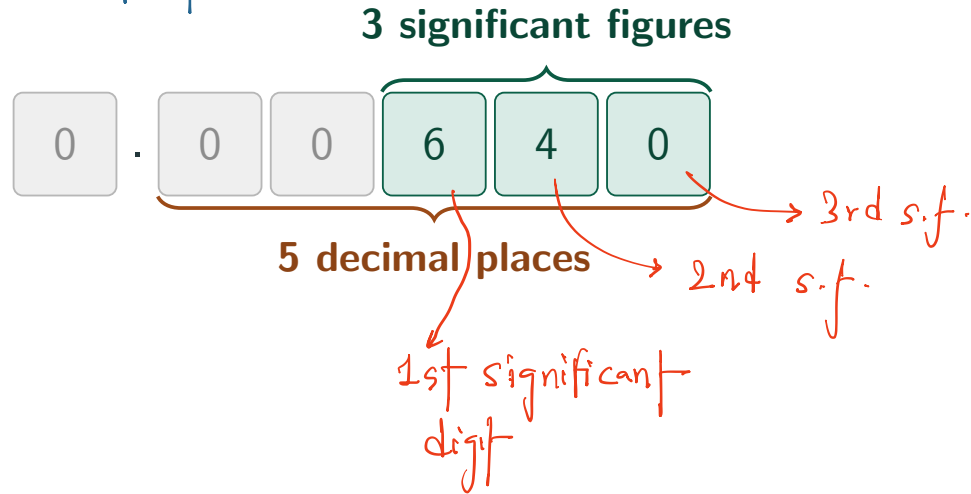
$$1,000,000 = 1 \times 10^6 \quad (1 \text{ s.f.}), \quad 1,079,587 = \underline{1.079587} \times 10^6 \quad (7 \text{ s.f.}).$$

↓
significant figures → same as significant digits

Decimal places $\neq$ significant figures

- **Decimal places** — count every digit *after* the point.
- **Significant figures** — meaningful digits, starting at the first non-zero one (let's not worry about leading zeros).

→ not related to decimal point



0.064045

7 decimal places

6 significant figures

Normalised scientific notation: $|x| = m \times 10^n$

Every non-zero number can be written

$$|x| = m \times 10^n, \quad 1 \leq m < 10, \quad n = \lfloor \log_{10} |x| \rfloor.$$

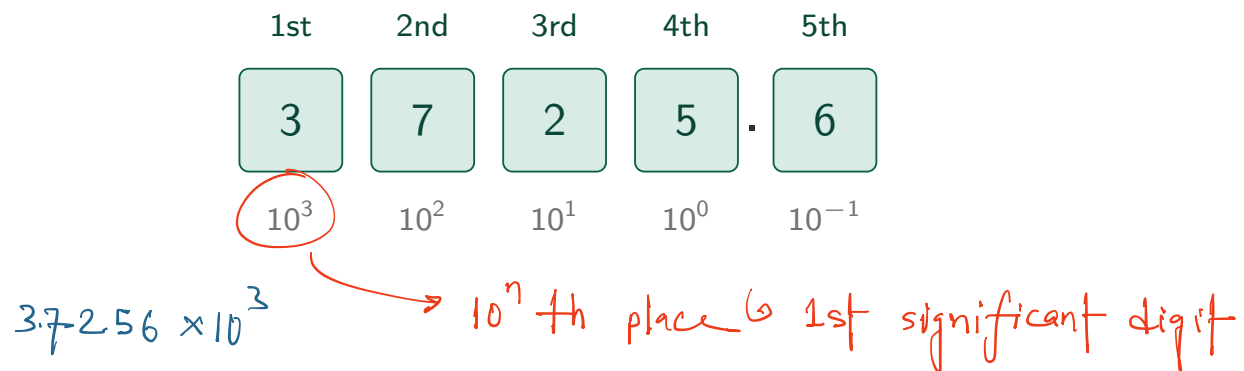
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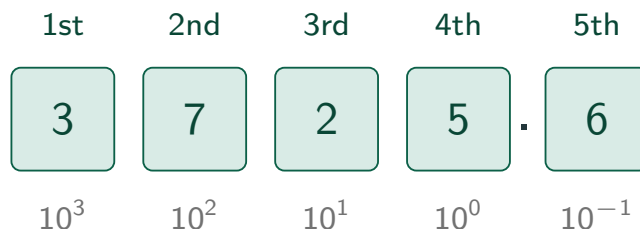


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$3725.6 = 3.7256 \times 10^3$, so $m = 3.7256$, $n = 3$; the leading 3 sits in the 10^3 (thousands) place

Where the M -th significant digit lives

$$\underbrace{10^n}_{1\text{st}}, \underbrace{10^{n-1}}_{2\text{nd}}, \underbrace{10^{n-2}}_{3\text{rd}}, \dots, \boxed{\underbrace{10^{n-M+1}}_{M\text{-th}}}.$$

Core fact

The M -th significant digit sits in the 10^{n-M+1} place.

Example. For $3725.6 = 3.7256 \times 10^3$ ($n = 3$), the 3rd significant digit (the 2) sits in the $10^{3-3+1} = 10^1$ place, the tens column.

The claim

If $\left| \frac{x - \hat{x}}{x} \right| \leq 0.5 \times 10^{2-M} \%$, then $\hat{x}$ is correct to at least M significant digits.

actual value র সাথে match হতে পারে।

$$\hat{x} \rightarrow \underline{5.683} 25 \times 10^{-5}, M=4$$

true result এর সাথে এই part aligned হবে
(confidently claim করছি)

$$x = 5.683 \underline{2}$$

predicted
① actual value

The claim

If $\left| \frac{x - \hat{x}}{x} \right| \leq 0.5 \times 10^{2-M} \%$, then $\hat{x}$ is correct to at least M significant digits.

Since “%” means “ $\times 10^{-2}$ ”, the same bound as a plain fraction is

$$0.5 \times 10^{2-M} \% = 0.5 \times 10^{2-M} \times 10^{-2} = \frac{1}{2} \times 10^{-M}.$$

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Example. For $M = 3$, the threshold: $0.5 \times 10^{-1} \% = 0.05 \%$ (i.e. $\frac{1}{2} \times 10^{-3} = 0.0005$)

Step 1: “Correct to M digits” meaning

Definition

$\hat{x}$ is correct to M significant digits when the error is **at most** *half a unit in the place of the M -th significant digit*:

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$$|x - \hat{x}| \rightarrow \text{error}$$

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$$\text{3rd s.f. 6th unit} = 1 \times 10^{0-3+1} = 0.01$$

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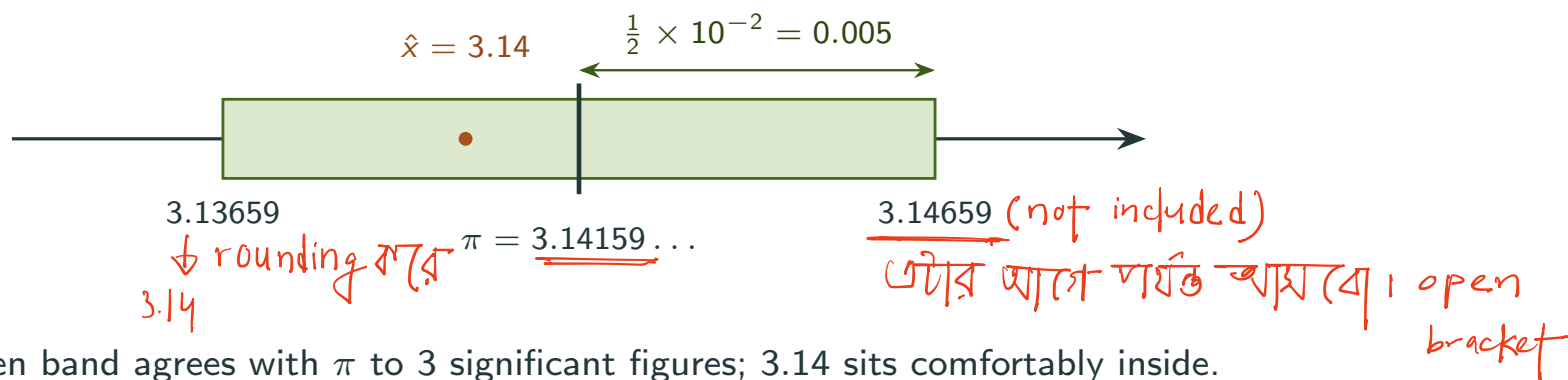
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so 3.14 is correct to 3 significant figures

Step 1: “Correct to M figures” meaning

“Correct to M figures” is an *absolute* window of half-width $\frac{1}{2} \times 10^{n-M+1}$ around x . For π with $M = 3$ that half-width is 0.005:



Any $\hat{x}$ inside the green band agrees with π to 3 significant figures; 3.14 sits comfortably inside.

Step 2: Equivalence of the two definitions

$$\text{Let } \left| \frac{x - \hat{x}}{x} \right| \leq \frac{1}{2} \times 10^{-M}$$

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$\nearrow |\varepsilon_n|$

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iff

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$\frac{1}{2} \times 10^{n-M+1}$ is exactly the window in Step 1, so $\hat{x}$ is correct to at least M significant digits. ■

Where do errors come from?

- **Modelling error.** Wrong assumptions.
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- Data / measurement error and programming mistakes.

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Inherent to numerical methods themselves *process এর কারণে inherently আছে*

Even with a perfect model and a perfect program, two errors remain:

(1) round-off error (2) truncation error.

device এর error

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Definition

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When I say - “I’ll be there in five minutes.”

Truncation error

Definition

The error from *truncating* (cutting short) a mathematical procedure.

Truncation error

→ maclaurin series দুইটি ম নিয়ে ছোট একটা অংশ নিলো

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The error from *truncating* (cutting short) a mathematical procedure.

Example. The Maclaurin series for $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ has infinitely many terms; keeping three gives

$$e^x \approx 1 + x + \frac{x^2}{2!}, \quad \text{truncation error} = e^x - \left(1 + x + \frac{x^2}{2!}\right) = \sum_{n=3}^{\infty} \frac{x^n}{n!}$$

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The discarded “tail” is the truncation error.

Controlling truncation with ϵ_a

Computing $|\epsilon_a|$ for $e^{1.2} = \sum_{k \geq 0} \frac{1.2^k}{k!}$ and stop once # significant digits ≥ 2 .
 $|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$ $\Rightarrow$ at least m significant digits.

n	S_n	E_a	$ \epsilon_a \%$	#s.d.
-----	-------	-------	------------------	-------

এই n সংখ্যক term এর sum

We approximate $e^{1.2}$ by summing the first n terms of the Maclaurin series:

$$S_n = \sum_{k=0}^{n-1} \frac{1.2^k}{k!}.$$

After each new term we check the *absolute relative approximate error*

$$|\epsilon_a| = \left| \frac{S_n - S_{n-1}}{S_n} \right| \times 100\%$$

and stop as soon as $|\epsilon_a| \leq 0.5\%$.

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n	S_n	E_a	$ \epsilon_a \%$	$\#$ s.d.
1	1.0000	—	—	—

S_1

Step 1

$$S_1 = 1 = 1.0000.$$

No previous value to compare against, so E_a and ϵ_a are undefined — we need a second estimate first.

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n	S_n	E_a	$ \epsilon_a \%$	$\#$ s.d.
1	1.0000	—	—	—
2	2.2000	1.2	54.546	0

Step 2 — add $\frac{1.2^1}{1!} = 1.2$.

$$S_2 = 1.0000 + 1.2 = 2.2000,$$

$$E_a = S_2 - S_1 = 1.2,$$

$$|\epsilon_a| = \left| \frac{1.2}{2.2000} \right| \times 100\% = 54.546\%.$$

$54.546\% > 50\%$ ($= 0.5 \times 10^2\%$) $\Rightarrow m = 0$: the first digit is not trustworthy yet.

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n	S_n	E_a	$ \epsilon_a \%$	#s.d.
1	1.0000	—	—	—
2	2.2000	1.2	54.546	0
3	2.9200	0.72	24.658	0

Step 3 — add $\frac{1.2^2}{2!} = 0.72$.

$$S_3 = 2.2000 + 0.72 = 2.9200,$$

$$E_a = 0.72,$$

$$|\epsilon_a| = \left| \frac{0.72}{2.9200} \right| \times 100\% = 24.658\%.$$

$$5\% < 24.658\% \leq 50\% \Rightarrow m = 0.$$

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n	S_n	E_a	$ \epsilon_a \%$	#s.d.
1	1.0000	—	—	—
2	2.2000	1.2	54.546	0
3	2.9200	0.72	24.658	0
4	3.2080	0.288	8.9776	0

Step 4 — add $\frac{1.2^3}{3!} = 0.288$.

$$S_4 = 2.9200 + 0.288 = 3.2080,$$

$$E_a = 0.288,$$

$$|\epsilon_a| = \left| \frac{0.288}{3.2080} \right| \times 100\% = 8.9776\%.$$

$$5\% < 8.9776\% \leq 50\% \Rightarrow m = 0.$$

Controlling truncation with ϵ_a

Computing $|\epsilon_a|$ for $e^{1.2} = \sum_{k \geq 0} \frac{1.2^k}{k!}$ and stop once # significant digits ≥ 2 .
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n	S_n	E_a	$ \epsilon_a \%$	#s.d.
1	1.0000	—	—	—
2	2.2000	1.2	54.546	0
3	2.9200	0.72	24.658	0
4	3.2080	0.288	8.9776	0
5	3.2944	0.0864	2.6226	1

Step 5 — add $\frac{1.2^4}{4!} = 0.0864$.

$$S_5 = 3.2080 + 0.0864 = 3.2944,$$

$$E_a = 0.0864,$$

$$|\epsilon_a| = \left| \frac{0.0864}{3.2944} \right| \times 100\% = 2.6226\%.$$

$0.5\% < 2.6226\% \leq 5\% \Rightarrow m = 1$: one digit earned.

3 is correct and guarantee দিতে পারছি।

4th-er পরন্তু গারান্টিয়াস না guarantee দিও।

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n	S_n	E_a	$ \epsilon_a \%$	$\#$ s.d.
1	1.0000	—	—	—
2	2.2000	1.2	54.546	0
3	2.9200	0.72	24.658	0
4	3.2080	0.288	8.9776	0
5	3.2944	0.0864	2.6226	1
6	3.3151	0.020736	0.62550	1

Step 6 — add $\frac{1.2^5}{5!} = 0.020736$.

$$S_6 = 3.2944 + 0.020736 = 3.3151,$$

$$E_a = 0.020736,$$

$$|\epsilon_a| = \left| \frac{0.020736}{3.3151} \right| \times 100\% = 0.62550\%.$$

$$0.5\% < 0.62550\% \leq 5\% \Rightarrow m = 1$$

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5	3.2944	0.0864	2.6226	1
6	3.3151	0.020736	0.62550	1
7	<u>3.3193</u>	0.0041472	0.12494	2

stop: $0.12494\% \leq 0.5\%$ at $n = 7$.

Step 7 — add $\frac{1.2^6}{6!} = 0.0041472$.

$$S_7 = 3.3151 + 0.0041472 = 3.3193,$$

$$E_a = 0.0041472,$$

$$|\epsilon_a| = \left| \frac{0.0041472}{3.3193} \right| \times 100\% = 0.12494\%.$$

$0.12494\% \leq 0.5\%$ ($= 0.5 \times 10^0\%$) $\Rightarrow$ $m = 2$. Bar cleared —
stop with at least **2** significant digits.

Truncation isn't only “chopping a series”

→ mathematical procedure (cut short)

To find the derivative, we define: $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$

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Example. For $f(x) = x^2$, the exact $f'(3) = 2(3) = 6$

With $\Delta x = 0.2$:

$$f'(3) \approx \frac{(3.2)^2 - (3)^2}{0.2} = \frac{10.24 - 9}{0.2} = 6.2,$$

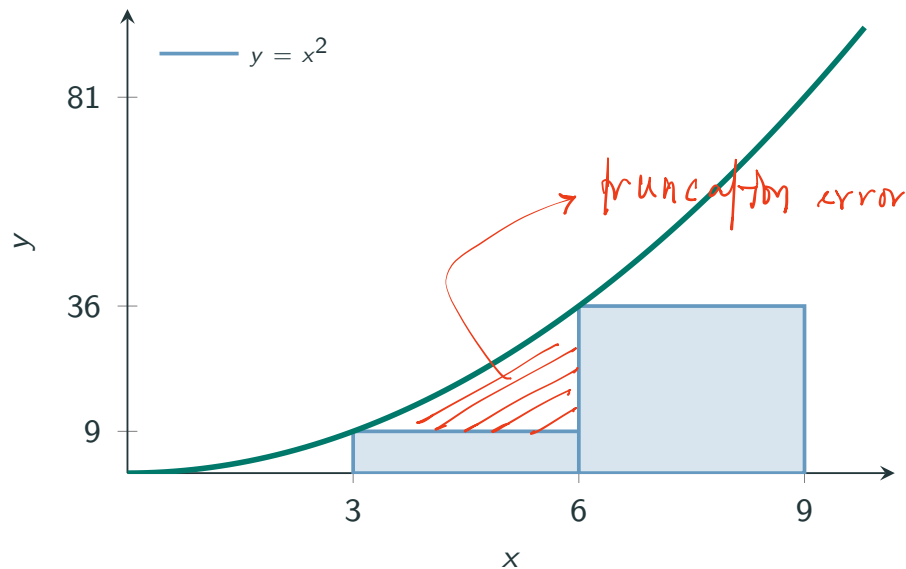
truncation error = $6 - 6.2 = -0.2$. 

Truncation in numerical integration

Exact area:

$$\int_3^9 x^2 dx = \left[\frac{x^3}{3} \right]_3^9 = \frac{729 - 27}{3} = 234.$$

Truncation in numerical integration



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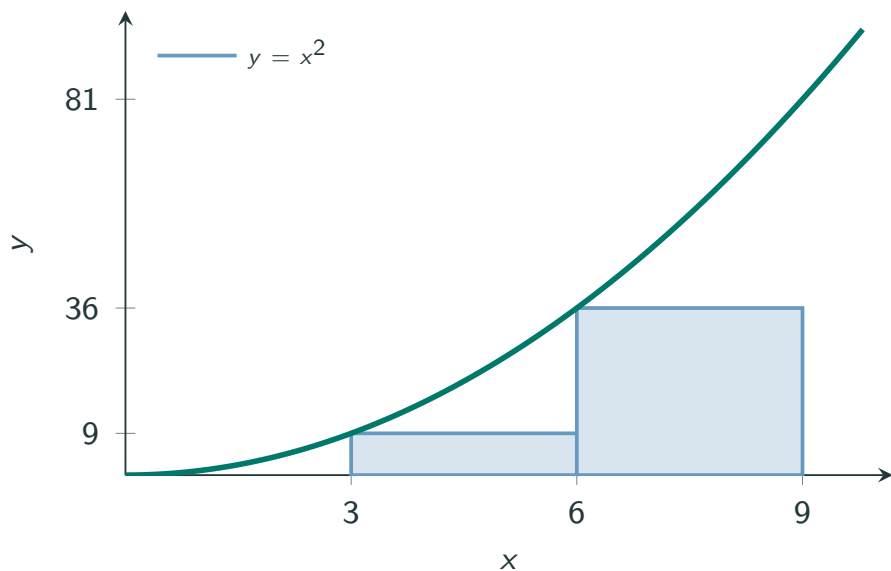
$$\int_3^9 x^2 dx = \left[\frac{x^3}{3} \right]_3^9 = \frac{729 - 27}{3} = 234.$$

Two left-endpoint rectangles (heights 9 and 36, width 3):

$$9 \cdot 3 + 36 \cdot 3 = 135.$$

$$\text{truncation error} = 234 - 135 = 99 (99/234 \approx 0.423).$$

Truncation in numerical integration



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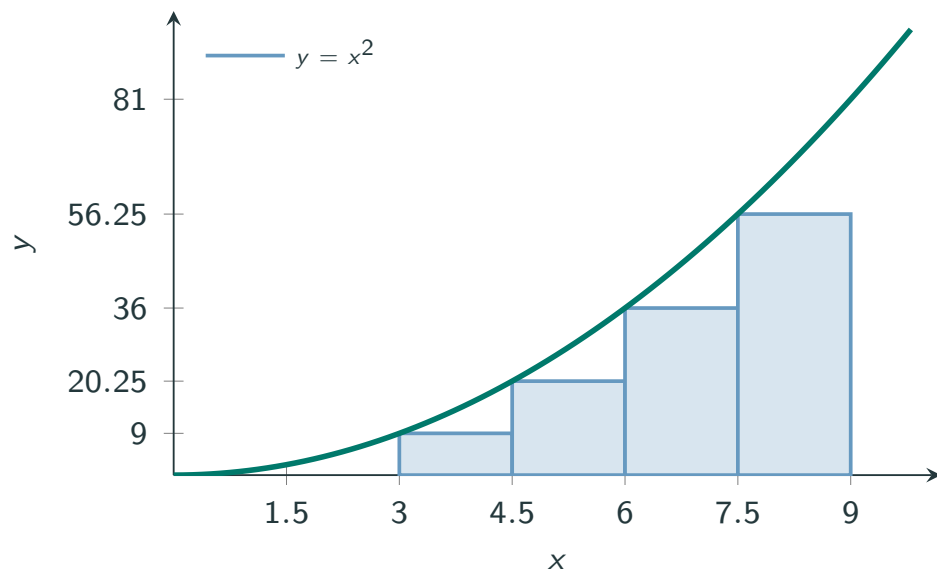
The white gap between curve and rectangles is the error.

42% error

Continuous process এর ফলাফল হিসেবে দেখানো।

Truncation in numerical integration

Continuous process এর ফাঁকে থেকে গোটানো। ওখন আরো কিছুইর সামনে গানো।



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$$9 \cdot 3 + 36 \cdot 3 = 135.$$

truncation error = $234 - 135 = 99$ ($99/234 \approx 0.423$).

The white gap between curve and rectangles is the error.

Four left-endpoint rectangles (width 1.5):

$$1.5 \times (9 + 20.25 + 36 + 56.25) = 182.25.$$

trunc. error = $234 - 182.25 = 51.75$ ($51.75/234 \approx 0.221$).

More rectangles $\Rightarrow$ smaller gap $\Rightarrow$ less error.

22% error

